

MACHINE LEARNING COURSE CONTENT

Duration- 6-8 Weeks

WEEK - 01

Introduction to Machine Learning

- What is Machine Learning?
- Types of ML: Supervised, Unsupervised, Reinforcement Learning
- Applications of ML in industry (healthcare, finance, IoT, etc.)
- Overview of ML tools: Python, Jupyter, Scikit-learn, TensorFlow
- Installing Python & Anaconda
- First program: Hello World ML using Scikit-learn

WEEK - 02

Python for Machine Learning

- Python basics (variables, loops, functions, classes)
- NumPy for numerical operations
- Pandas for data manipulation
- Matplotlib & Seaborn for data visualization
- Exploratory Data Analysis (EDA) with real datasets

WEEK - 03

Supervised Learning - Regression

- Linear Regression: theory and implementation
- Polynomial Regression
- Evaluation metrics: MSE, RMSE, R^2 Score
- Practice: Predict housing prices or sales data

WEEK - 04

Supervised Learning - Classification

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Decision Trees and Random Forests
- Confusion Matrix, Precision, Recall, F1-Score
- Practice: Email spam detection or Iris classification

WEEK - 05

Unsupervised Learning

- Clustering: K-Means, Hierarchical Clustering
- Dimensionality Reduction: PCA
- Applications: Customer segmentation, Anomaly detection
- Practice: Segmenting mall customers or patterns in IoT data

WEEK - 06

Model Evaluation & Optimization

- Cross-validation
- Hyperparameter tuning: Grid Search, Random Search
- Bias-Variance trade-off
- Overfitting and underfitting
- Practice: Train/test split & cross-validation in real project

OPTIONAL ADVANCED WEEKS

WEEK - 07

Neural Networks & Deep Learning

- Introduction to Neural Networks
- Perceptron, Activation functions
- Using Keras/TensorFlow for simple neural networks
- Practice: Handwritten digit recognition using MNIST dataset

WEEK - 08

Final Project & Presentation

- Choose and implement an ML project (end-to-end)
- Data collection, preprocessing, modeling, evaluation
- Report writing and presentation
- Peer review and feedback

WEEK - 09

Course Deliverables

- Lecture Slides
- Notebooks for examples
- Practice datasets
- Weekly assignments & quizzes
- Final mini-project

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PASSION FOR INTEGRATED SOLUTIONS